

Amritaraj Nair

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EDUCATION

Texas A&M University — B.S. Computer Science Honors, Minor in Mathematics

Expected May 2029

Relevant Coursework: Data Structures & Algorithms, C++ Design, Discrete Structures, Computer Architecture and Assembly, Python Programming

EXPERIENCE

Matic | *Software Engineer Intern*

Jun 2026 – Aug 2026

- Shipped **Pulsematic**, an AI inbox-triage and auto-drafting system for **athenaOne** and other **EHR inboxes** — saved physicians **90+ min/provider daily** (3 more patients/day).
- Built a **multi-agent orchestrator** on **GCP + Vertex AI** to route, prioritize, and optimize drafts across message types, reducing manual edits.
- Automated **CI/CD with Google Cloud Build triggers**, cutting release turnaround across services.
- Engineered a **PHI-compliant data pipeline on Firestore** with strict access controls.

MagNet Agents | *Software Engineer Intern*

May 2025 – Aug 2025

- Built **data pipelines and web-scraping systems** to extract and structure legal lead data from public sources.
- Developed **LLM-powered outreach** generating personalized messaging and legal-opportunity summaries, lifting engagement and conversion.
- Built **backend lead-matching and tracking** connecting lawyer profiles to prospects; supported a Cornell-backed platform at **\$50K+ ARR**.

AI Modeling for Bio-Economic Systems | *Undergraduate Research*

Aug 2025 – Present

- Optimized livestock nutrition under economic and environmental constraints using **agent-based modeling (Mesa)** and **ML**.
- Migrated legacy **R workflows to Python**, improving reproducibility and scalability.
- Built cross-platform simulations in **Python, Java, and NetLogo** to evaluate sustainability and production outcomes.

Maroon Fund — Scholars of Finance | *Quantitative Developer*

Feb 2025 – Present

- Built **quantitative models and screening frameworks** for equity research across a **\$70K+ student-managed fund**.
- Engineered **data pipelines and valuation tools** on financial datasets, speeding analysis.
- Automated **research workflows** powering stock-pitch and portfolio decisions.

PROJECTS

ClinicalHours (clinicalhours.org) | *CTO*

Dec 2025 – Present

- Built an **AI email/call automation system (Gmail API, GoHighLevel)** acting as a virtual receptionist to schedule clinic meetings — placed **top 10% of the Spring 2026 Y Combinator batch**.
- Integrated **MapBox, Google APIs, and Resend API** for geolocation clinic discovery and automated communication pipelines.
- Processed large-scale **U.S. hospital datasets** into production-ready formats in Python.
- Shipped AI resume and application tools (**OpenAI, Gemini**); partnered with clinics including **BCS Free Health Clinic**.

Thorp (Thorp on App Store · thorp-trade.vercel.app) | *Founder & Solo Developer*

Mar 2026 – Present

- Launched the **TikTok of finance** for next-gen traders: an iOS and web app that turns finance news into a **swipeable short-form feed**; invited by **YC partner Ryan Choi** to the **YC Startup Intern Expo**.
- Designed a **fault-tolerant distributed systems layer** across 5 providers on stateless serverless workers, making market data **10× faster** (1.72s → 0.17s p50) with **50× fewer API calls**.
- Engineered a **backtesting engine** (9 indicators, expectancy, drawdown) that exposes overfitting, plus a **visual strategy builder with Python export** and a **from-scratch SVG charting engine**.
- Built the **social layer** (posts, follows, paper-trading P&L) on PostgreSQL; now building **parallel agent orchestration** to generate and score feed content.

Shot Sensei | *Co-Founder & Lead Developer*

Mar 2026

- Built a **computer-vision pickleball platform**; won the Startup Ready Award at **Hook'em Hacks 2026 (UT Austin)**.
- Shipped a **real-time mode** (play vs AI bot) plus a **training mode** using **Gemini and ElevenLabs** to coach all strokes (serve, volley, forehand, backhand).
- Implemented **pose/shot detection** with **OpenCV and YOLOv8**; **Supabase** for win/loss records and shot-level analytics.
- Invited to **McCombs School of Business (UT Austin)** to pitch to **Pear VC** (Khalil Fuller).

Prophecy (amritnair.github.io/prophecy) | **HackMIT 2026 Track Finalist** | *Lead Developer*

Sep 2026

- Built an **MCP-powered developer tool** mapping codebase dependencies, identifying change blast radius, and delivering targeted context to AI coding agents.
- Engineered **collaborative agent workflows** enabling AI coding assistants to share file-level findings, track overlapping work, and reduce redundant repository exploration.
- Developed **automated change-risk analysis** using dependency graphs and concurrent agent sessions to surface potential breaking changes before commits.

TECHNICAL SKILLS & HONORS

Languages: Python, C++, Java, TypeScript, SQL

Frameworks: React, Flask, FastAPI, Pandas, NumPy, Matplotlib

Tools: Git, Docker, AWS EC2/IAM, Google Cloud, Vercel, PostgreSQL, Supabase, CI/CD

Awards: YC Startup Intern Expo (recruited by YC partner) · UT Austin Hook'em Hacks 2026 **Winner** (Most Startup Ready + Multimodal Track) ·

TidalTAMU Google Gemini Track **1st Place** (2026) · Outstanding Undergraduate Researcher Award · President's Endowed Scholar